

DIFFERENTIAL AMPLIFIER

CIRCUIT 1

Aim:

"Design and analysis of the MOS differential amplifier circuit for the following specifications."

Drain Supply Voltage (VDD): 0.9V

Source Supply Voltage (VSS): -0.9V

Maximum Power Dissipation (P): $\leq 1.8\text{mW}$

Input Common Mode (VinCM): 0V

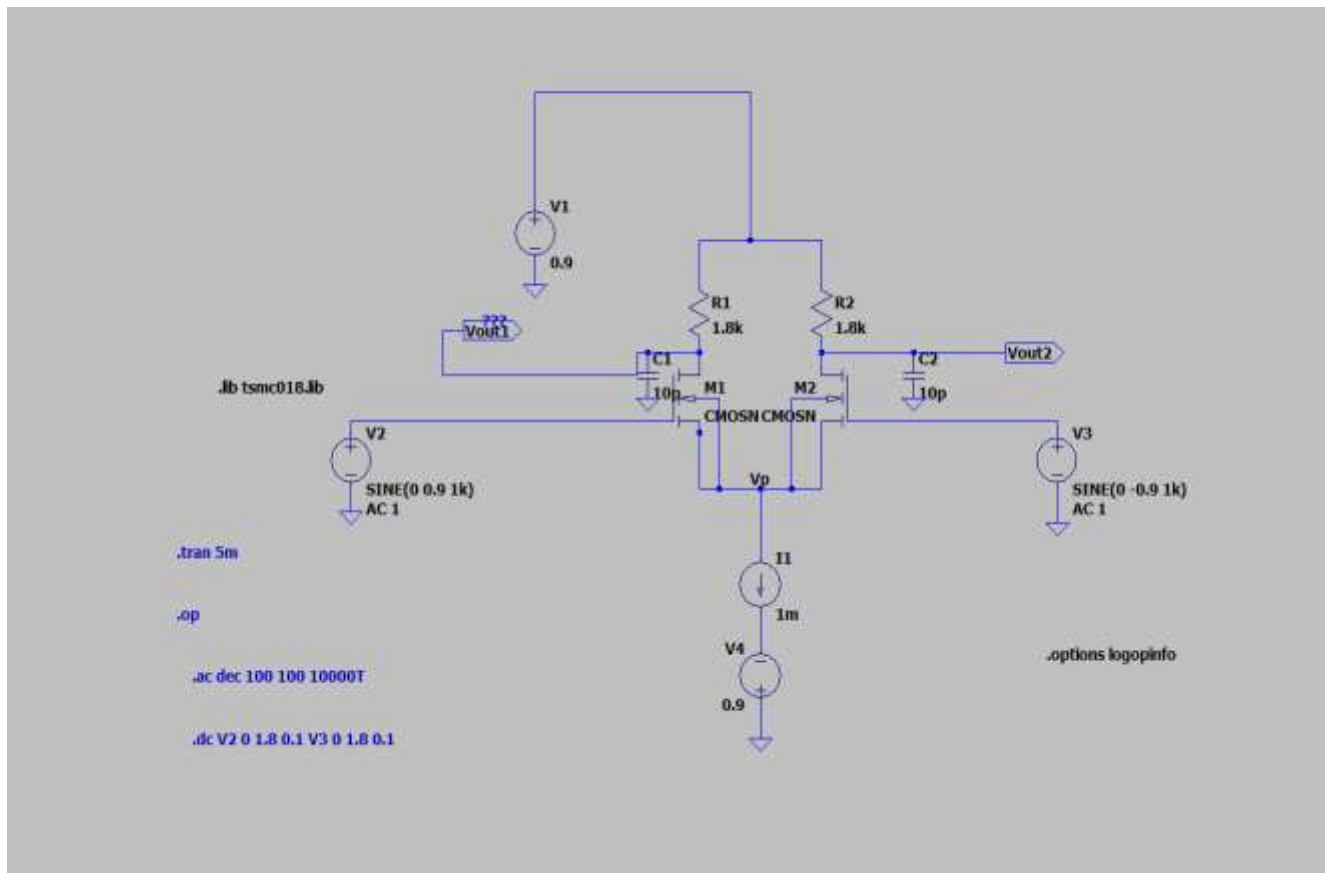
Output Common Mode (VoCM): 0V

Tail Node Voltage (VP): -0.7V

Load Capacitance (CL): 10pF

length : 480nm; $\epsilon_r = 3.9$ $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$ $t_{ox} = 4.1 \times 10^{-9} \text{ m}$; $\mu_n = 273.809 \text{ cm}^2/\text{Vs}$; $\mu_p = 115.689 \text{ cm}^2/\text{Vs}$

Circuit Diagram:



1.DC analysis:

In the ideal balanced state ($V_{in1} = V_{in2}$), the tail current I_{SS} (labeled I1 in your schematic) splits equally between the two transistors (M1 and M2).

$$I_{D1} = I_{D2} = \frac{I_{SS}}{2}$$

$$I_{SS} = 1\text{mA}$$

$$V_{out(DC)} = V_{DD} - (I_D \cdot R_d)$$

$$0 = 0.9 - R_d \cdot 0.5\text{mA}$$

$$R_d = 1.8\text{k}$$

Verification for circuit:

$$V = 0.9\text{V}$$

$$I_D = 0.5\text{mA}$$

$$R_D = 1.8\text{k}$$

$$V_{out(DC)} = 0$$

NMOS width calculation (for both M1 and M2)

Oxide Capacitance (Cox) $Cox = \epsilon_r \epsilon_0 / t_{ox}$ $Cox = (3.9 \times 8.854 \times 10^{-12}) / (4.1 \times 10^{-9})$ $Cox = 8.42 \times 10^{-3} \text{ F/m}^2$

$V_{GS} - V_{TH} = 0.7 - 0.36 = 0.34\text{V}$

Drain current equation: $I_{D/2} = (1/2) \mu_n Cox (W/L) (VOV)^2$ Rearranging: $W_n = (I_{D/2} L) / (\mu_n Cox (VOV)^2)$ Substitute values: $W_n = (1 \times 10^{-3} \times 480 \times 10^{-9}) / (273.809 \times 10^{-4} \times 8.42 \times 10^{-3} \times (0.34)^2)$ $W_n \approx 18 \mu\text{m}$

$W_n = 19.9 \mu\text{m} \rightarrow I_d = 0.5\text{mA}$

Vincmin and Vincmax calculation:

Input Common-Mode Range (ICMR) Analysis:

The ICMR defines the allowable voltage range at the gates where all transistors remain in the Saturation Region.

1. Maximum Input Voltage (Vin,max)

To prevent M1 / M2 from entering the triode region, the gate voltage must stay below the drain voltage plus one threshold voltage (Vth):

Formula: $V_{in,max} = V_{DD} - (I_D * R_d) + V_{th}$

Calculation (Assuming $V_{th} \approx 0.36\text{V}$):

$V_{in,max} = 0.9\text{V} - (0.5\text{mA} * 1.8\text{k}\Omega) + 0.36\text{V}$

$V_{in,max} = 0.9\text{V} - 0.9\text{V} + 0.36\text{V}$

$V_{in,max} = 0.36\text{V}$

2. Minimum Input Voltage (Vin,min)

To ensure the tail current source I1 has enough "headroom" ($V_{DS,sat} \approx 0.2\text{V}$) to stay active:

Formula: $V_{in,min} = V_{SS} + V_{DS,sat} + V_{GS}$

Calculation (Assuming $V_{GS} \approx 0.6\text{V}$):

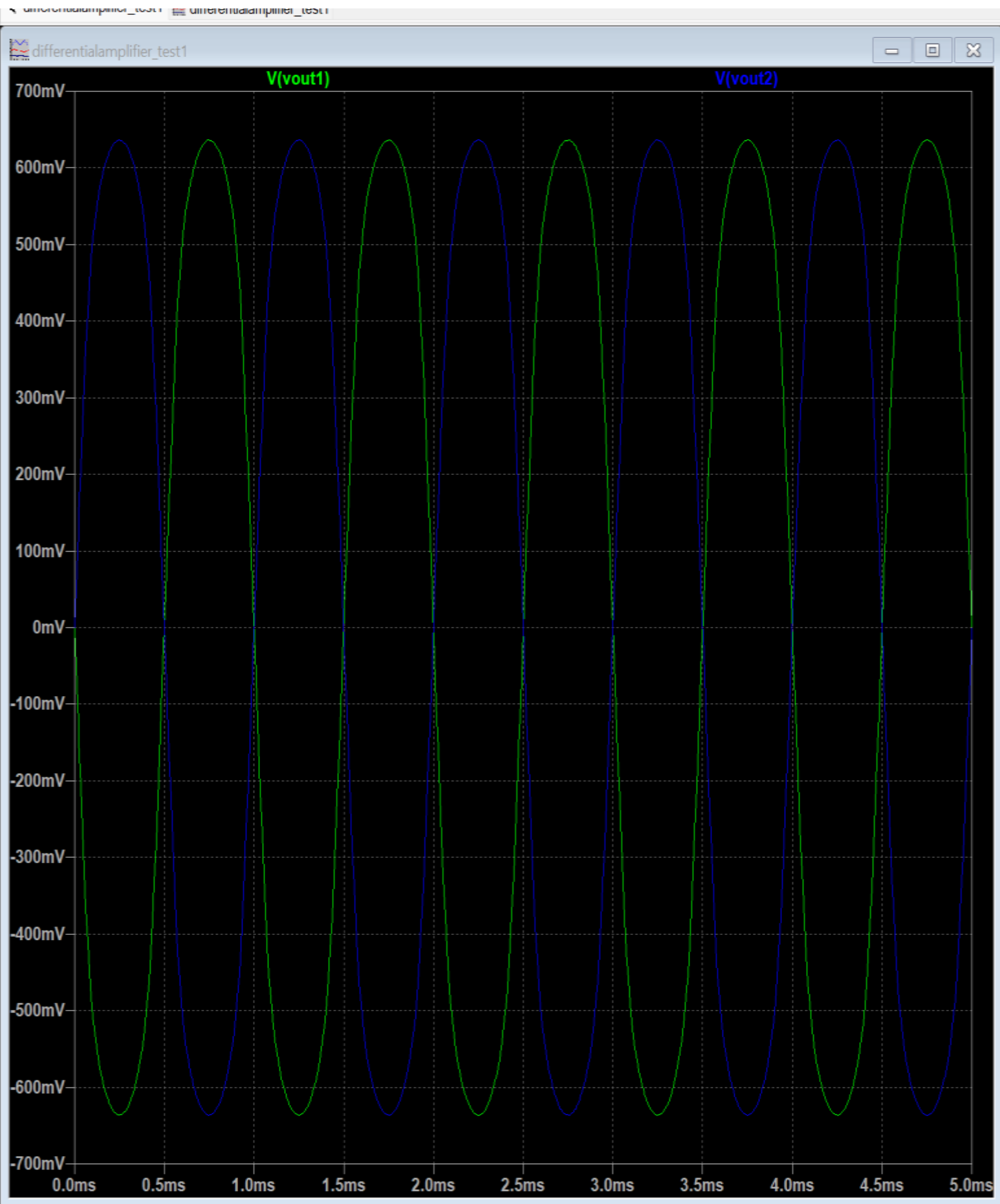
$V_{in,min} = -0.9\text{V} + 0.2\text{V} + 0.6\text{V}$

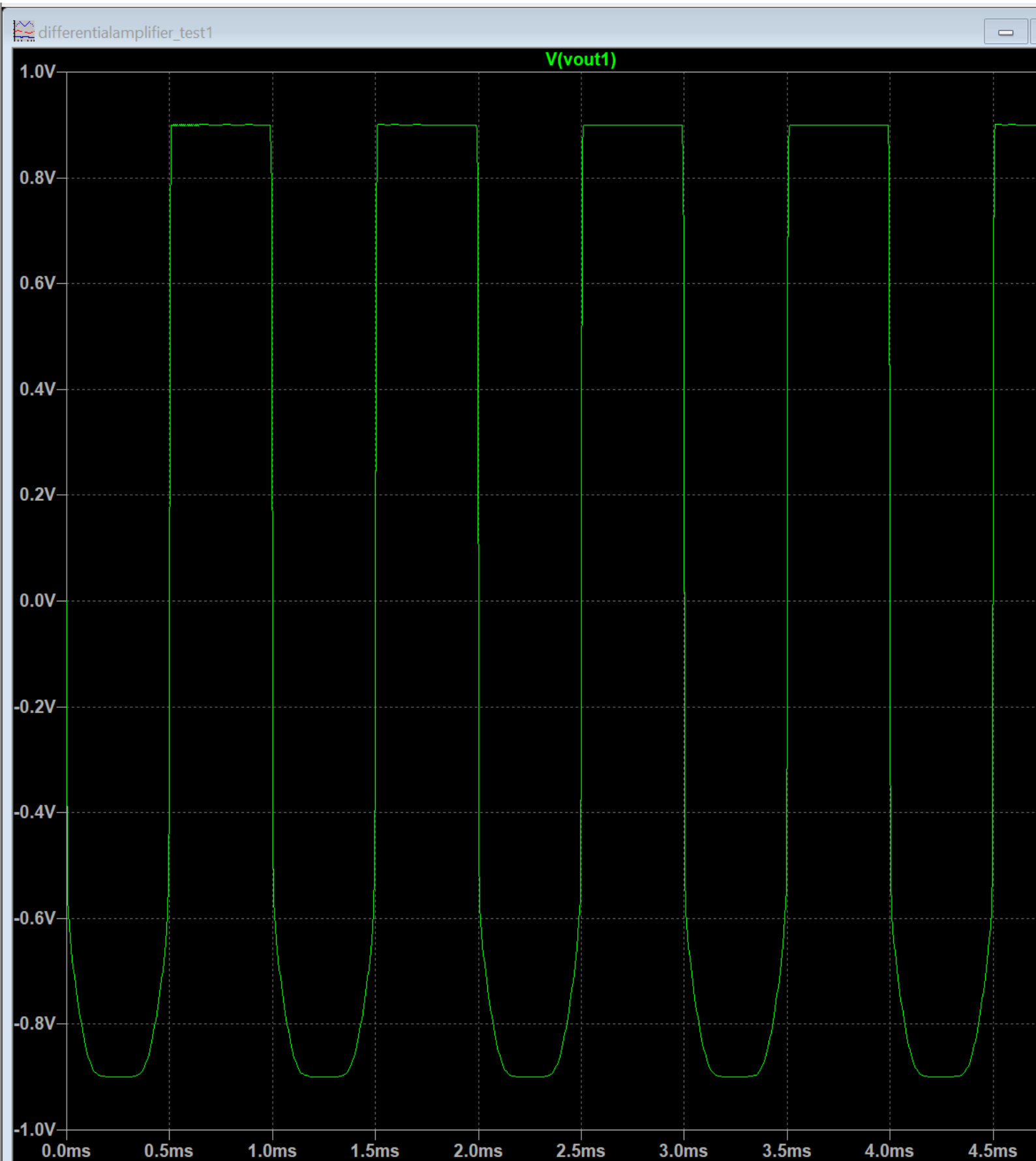
$V_{in,min} = -0.1\text{V}$

ICMR Summary Table:

Boundary	Condition	Values
Upper Limit (Vin,max)	Saturation of M1,M2	+0.4V
Lower Limit(Vin,min)	Tail Source Headroom	-0.1V

For linear amplification, input common-mode signal must stay between -0.9V and +0.36V.





if we increase the vin values then graph will be clipped(square form)

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C:\Users\Vanshikha Muthanna C\OneDrive\Desktop\180nm\experiment 4.net
--- Operating Point ---
V(n001):      0.9          voltage
V(n002):      0           voltage
V(n003):      0           voltage
V(n004):     -0.9          voltage
V(vout1):     7.09304e-16  voltage
V(vout2):     8.88178e-16  voltage
V(vp):       -0.724145    voltage
I(C1):        7.09304e-39  device_current
I(C2):        8.88178e-39  device_current
I(I1):        0.001        device_current
I(R1):        0.0005        device_current
I(R2):        0.0005        device_current
I(V1):       -0.001        device_current
I(V2):        0           device_current
I(V3):        0           device_current
I(V4):       -0.001        device_current
Ib(M1):      -7.34145e-13  device_current
Ib(M2):      -7.34145e-13  device_current
Id(M1):       0.000500001   device_current
Id(M2):       0.000500001   device_current
Ig(M1):       0           device_current
Ig(M2):       0           device_current
Is(M1):      -0.000500001   device_current
Is(M2):      -0.000500001   device_current

```

Vocm max and Vocm min calculation:

Vocmax=VDD = 0.9v

Vocmin = VDD-IDRD = 0v

Small-Signal Differential Gain (Ad) Analysis

The Differential Gain (Ad) defines how much the difference between the two input signals (Vin1 - Vin2) is amplified at the output (Vout1 - Vout2).

1. Fundamental Formula

For a MOSFET differential pair with resistive loads, the gain is given by:

$$Ad = -gm * Rd$$

Where:

g_m : Transconductance of the NMOS transistors (M1, M2).

R_d : Drain resistance (1.8k Ω).

2. Calculating Transconductance (g_m)

To get an accurate gain, we first find g_m . Using the branch current ($I_D = 0.5\text{mA}$):

Formula: $g_m = \sqrt{2 \cdot \mu_n \cdot C_{ox} \cdot (W/L) \cdot I_D}$ or $g_m = (2 \cdot I_D) / (V_{GS} - V_{th})$

Assuming typical 0.18 μm parameters ($V_{ov} \approx 0.34\text{V}$)

$$g_m = (2 \cdot 0.5\text{mA}) / 0.34\text{V} = 2.94\text{mS (mA/V)}$$

3. Final Gain Calculation

Plugging the values into the gain equation:

$$A_d = -(2.94\text{mS}) \cdot (1.8\text{k}\Omega)$$

$$A_d = -5.29 \text{ V/V}^{**}$$

In Decibels (dB): $^{**} \text{Gain(dB)} = 20 \cdot \log_{10}(5.29) \approx 14.46\text{dB}$

Gain Summary Table:

Parameter	Symbol	Estimated Value
Transconductance	g_m	5mS
Load Resistance	R_d	1.8k Ω
Differential gain	A_d	9V/V(14.469dB)
Output Type	Differential	$V_{out1} - V_{out2}$

practical gain

Measured peak-to-peak value Measured peak-to-peak values:

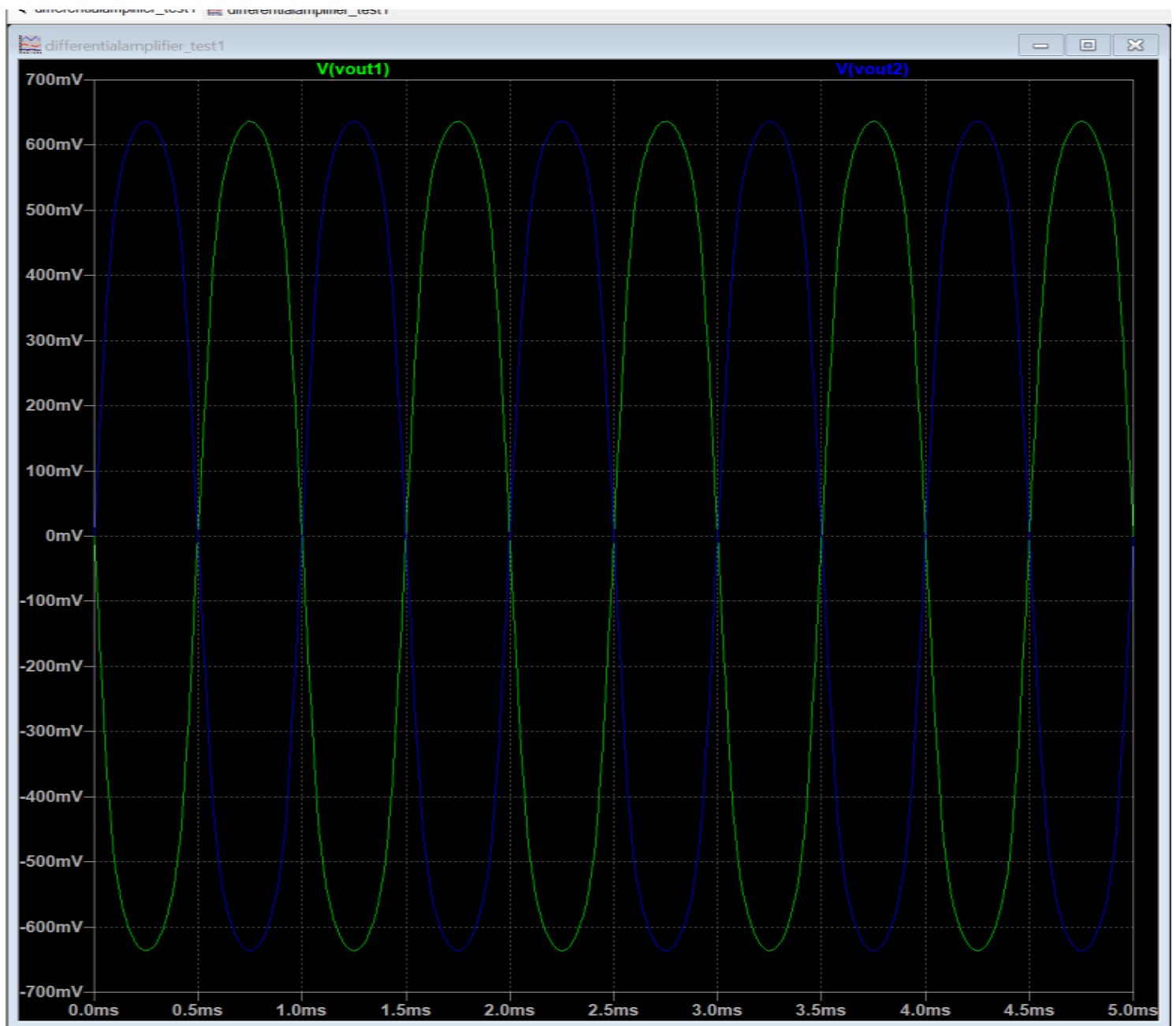
- **Input Voltage:** $V_{in(p-p)} = 100\text{mV} - (-100\text{mV}) = 200\text{mV}$
- **Output Voltage:** $V_{out(p-p)} = 602\text{mV} - (-602\text{mV}) = 1204\text{mV}$

- $AV = v_{outp-p}/v_{inp-p}$
- $1204/200 === 6.02$
- $V(DB) = 20 \cdot \log(6.02) = 15.59$

$v_{id} < \sqrt{2} V_{ov}$

When the differential input voltage V_{id} (the difference between your two inputs V_2 and V_3) is less than $\sqrt{2} \cdot V_{ov}$ (where V_{ov} is the overdrive voltage $V_{GS} - V_{th}$), the circuit is operating in its Linear **Operation for $V_{id} < \sqrt{2} V_{ov}$ (Linear Region)**

When the differential input V_{id} is small, both transistors (M_1 and M_2) remain **ON** and conduct a portion of the tail current. This is the intended Region of the amplifier.



1. Current Steering Behavior

In this range, the tail current I_{SS} (1mA) is shared between both branches. The current in each transistor changes linearly with the input:

- $I_{D1} \approx \frac{I_{SS}}{2} + \frac{g_m V_{id}}{2}$
- $I_{D2} \approx \frac{I_{SS}}{2} - \frac{g_m V_{id}}{2}$

As V_{in1} increases, M_1 takes more of the 1mA current, and M_2 takes less, but **neither transistor is cut off yet**.

2. Output Voltage Effect

Since the currents are changing linearly, the output voltage at V_{out1} and V_{out2} will track the input signal accurately without significant distortion.

Differential Output Formula: $V_{out1} - V_{out2} = -g_m \cdot R_d \cdot V_{id}$

For your circuit:

- The output will be a **sine wave** (if the input is a sine wave).
- The gain remains constant at approximately **5.29 V/V**.
- The output signal will be "clean" and follow the shape of the input.

Beyond this point ($V_{id} > \sqrt{2}V_{ov}$), the amplifier **saturates**. The output will "flat-top" or clip because the current cannot increase further than the 1mA provided by the tail source.

Operation Summary:

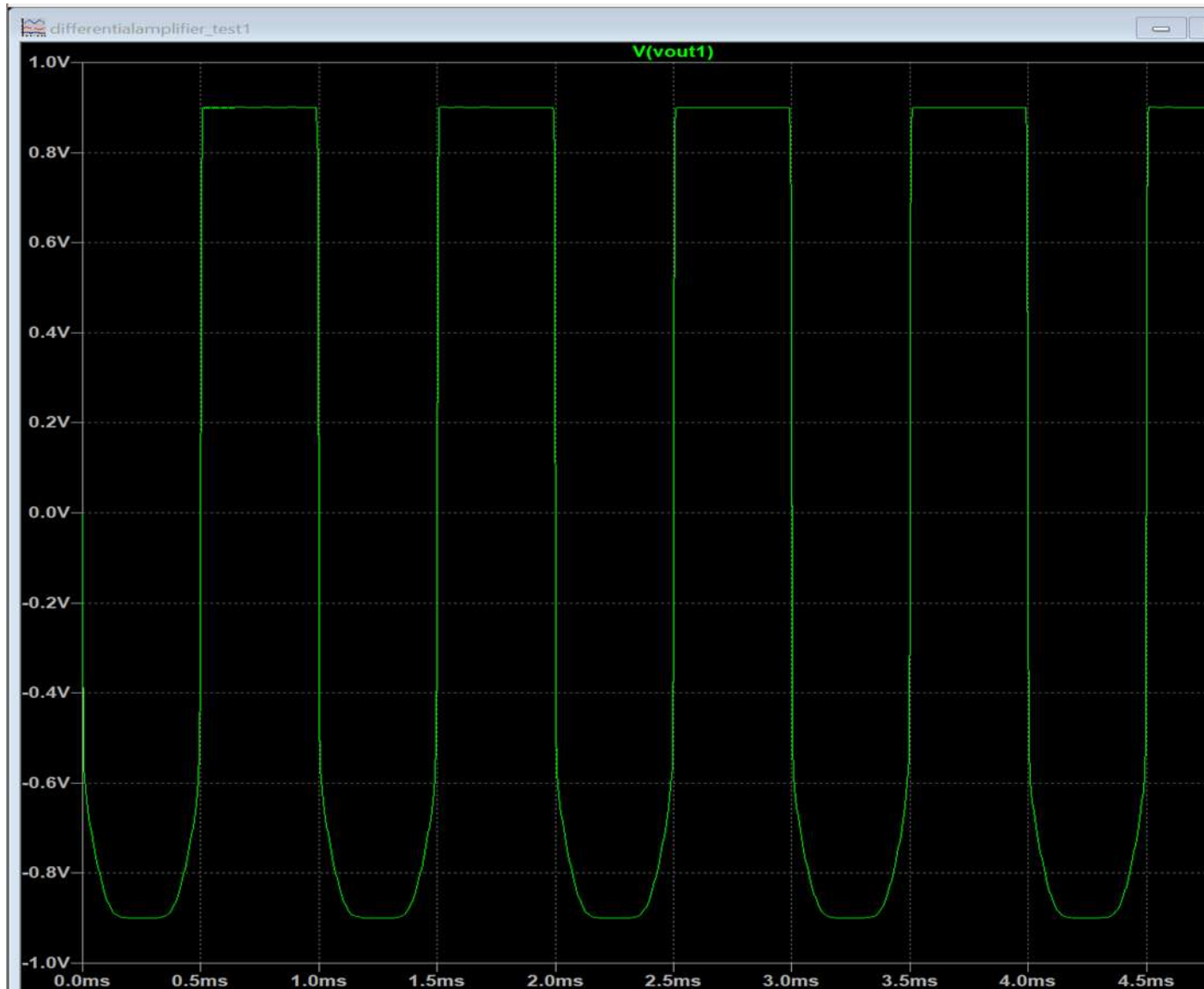
Operation Summary

Condition	Transistor Status	Output Behaviour
$V_{id} < \sqrt{2}V_{ov}$	Both M1, M2 are ON	Linear Amplification (No Clipping)
$V_{id} = \sqrt{2}V_{ov}$	One is maxed, one is at 0	Edge of Saturation
$V_{id} > \sqrt{2}V_{ov}$	One is OFF	Output Clipping (Distortion)

vid>root2 Vov

Operation for $V_{id} > \sqrt{2}V_{ov}$ (Current Steering/Clipping)

When the input difference is larger than $\sqrt{2}V_{ov}$, the differential pair can no longer amplify the signal linearly. The tail current is completely



1. Transistor States (The "Switch" Effect)

In this condition:

- **One transistor (M_1 or M_2)** carries the **entire 1mA** tail current.
- **The other transistor** is completely **OFF** (Cutoff), carrying 0mA.

2. Output Voltage Saturation (Clipping)

Since the current in the branches can no longer increase (it is limited by the 1mA tail source), the output voltage reaches its maximum possible "swing" and stays there. This is known as **Clipping**.

Maximum Output Levels:

- $V_{out,min}: V_{DD} - (I_{SS} \cdot R_d) = 0.9V - (1mA \cdot 1.8k\Omega) = \mathbf{0V}$
- $V_{out,max}: V_{DD} - (0 \cdot R_d) = \mathbf{0.9V}$

The differential output ($V_{out1} - V_{out2}$) will flat-line at $\pm 0.9V$.

3. Visual Changes in the Output Waveform

If you run a **Transient Analysis (.tran)**, you will see the following:

- **Shape Distortion:** A sine wave input will turn into a **Square-like wave** with "flat tops."
- **Gain Drop:** The "Small-Signal Gain" (A_v) effectively drops to zero at the peaks because $dV_{out}/dV_{in} = 0$.
- **Harmonic Distortion:** The output will contain high-frequency harmonics because it is no longer a pure sine wave.

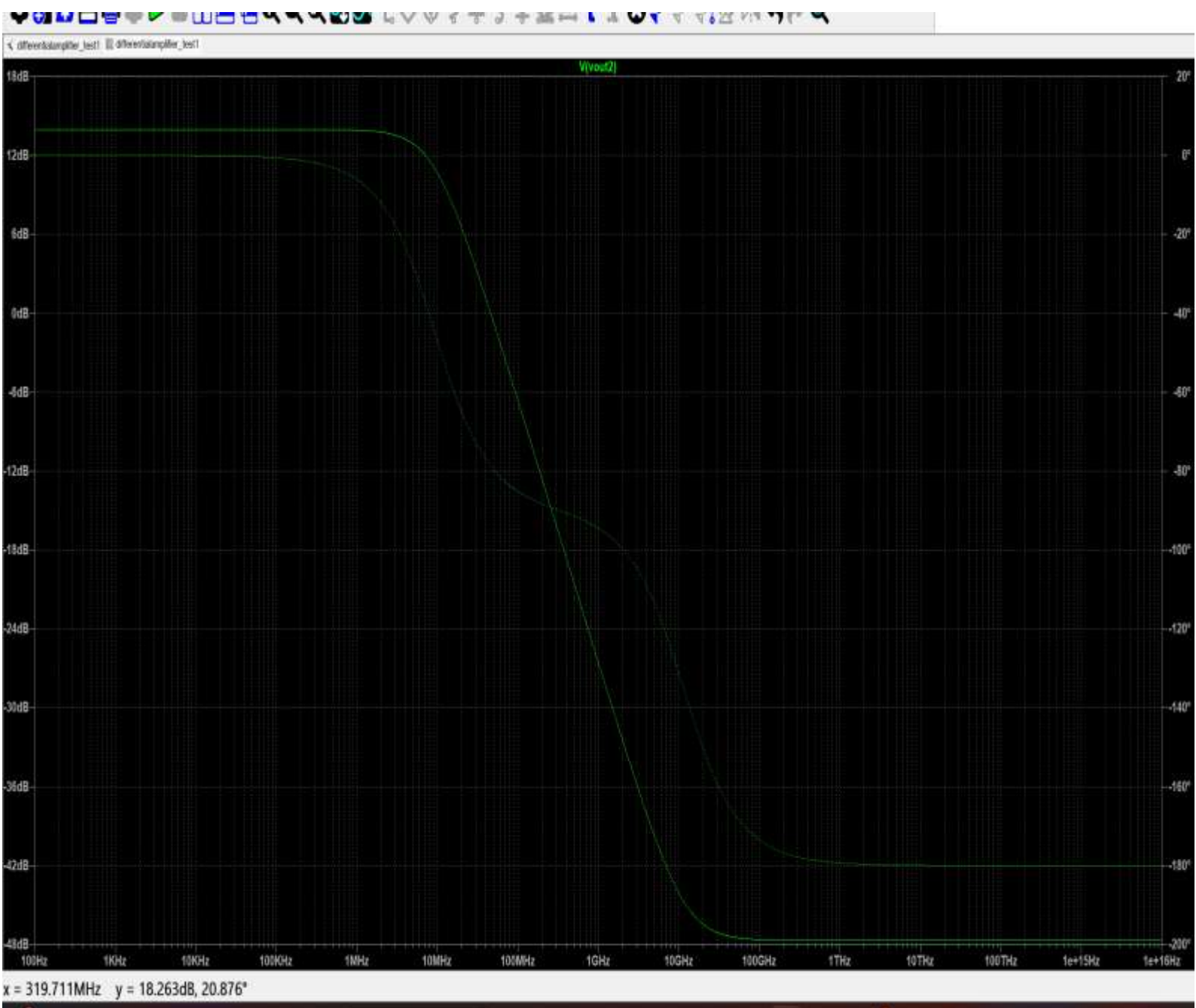
Comparison Table

Feature	Linear ($V_{id} < \sqrt{2}V_{ov}$)	Saturated ($V_{id} > \sqrt{2}V_{ov}$)
Current I_{D1}, I_{D2}	Both transistors share current	One is 1mA, one is 0mA
Output Waveform	Clean Sine Wave	Clipped / Flat-topped
Transconductance	Constant g_m	g_m drops to 0

Feature	Linear ($V_{id} < \sqrt{2}V_{ov}$)	Saturated ($V_{id} > \sqrt{2}V_{ov}$)
Function	Linear Amplifier	Comparator / Switch

Your Simulation: Since your input V_2 and V_3 have a 0.9V amplitude, your total V_{id} is **1.8V peak-to-peak**. This is massive compared to a typical $\sqrt{2}V_{ov} \approx 0.3V$. You will see extreme clipping in LTspice results.

AC analysis:



$$AV = 14.33\text{db}$$

$$AV-3 = 13.33-3 = 10.33$$

$$FL = 0\text{HZ}$$

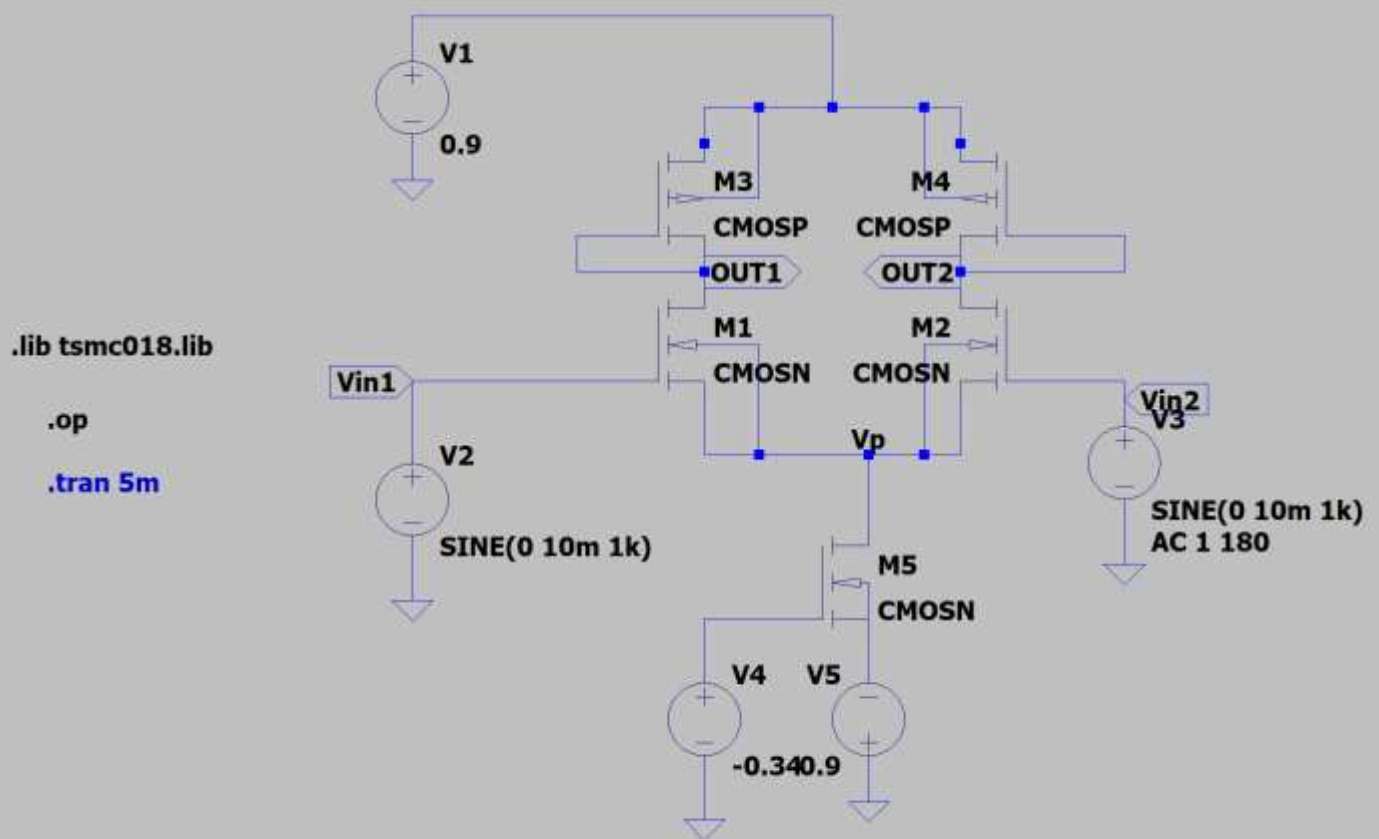
$$FH = 7.21\text{HZ}$$

$$BW = 7.21$$

$$UGB = AV * BW$$

$$UGB = 74.47$$

CIRCUIT 2



"Active load functionality is provided by the M_3 – M_4 PMOS current mirror. M_3 serves as the diode-connected reference, while M_4 mirrors the current to the opposite branch, achieving a much higher output resistance than possible with standard resistors."

GIVEN PARAMETERS

Design Specifications

Parameter	Symbol	Value
Supply Voltage (Positive)	V_{DD}	+0.9V
Supply Voltage (Negative)	V_{SS}	−0.9V
Power Constraint	P	= 1.8mW
Channel Length	L_n	480nm
Input Common-Mode Voltage	$V_{in,CM}$	0V
Output Common-Mode Voltage	$V_{O,CM}$	0V
Tail Node Voltage	V_p	−0.7V
Load Capacitance	C_L	10pF
Threshold Voltage	V_T	$\approx 0.36V$

Power Analysis

The total power consumed by the differential amplifier is calculated using the total voltage swing and the tail current (I_{SS}):

$$P = (V_{DD} - V_{SS}) \cdot I_{SS}$$

1. Total Supply Voltage: $V_{DD} - V_{SS} = 0.9V - (-0.9V) = 1.8V$

2. Maximum Allowed Power: Given the constraint $P \leq 1.8mW$: $1.8V \cdot I_{SS} \leq 1.8 \times 10^{-3}W$

3. Maximum Tail Current: $I_{SS} = 1mA$

Quiescent Current Distribution

Under zero differential input ($V_{in,diff} = 0$), the tail current splits equally between the two branches of the differential pair:

$$I_{D1} = I_{D2} = \frac{I_{SS}}{2}$$

Substituting the previously calculated I_{SS} :

$$I_{D1} = I_{D2} = \frac{1mA}{2}$$

$$I_{D1} = I_{D2} = 0.5mA$$

PMOS Operating Conditions

1. Terminal Voltages

- **Source (V_S):** Connected to $V_{DD} = 0.9V$
- **Drain (V_D):** 0V

2. Source-Drain Voltage Calculation The source-drain voltage (V_{SD}) is determined by the difference between the source and drain potentials:

$$\begin{aligned}V_{SD} &= V_{DD} - V_D \\V_{SD} &= 0.9V - 0V \\V_{SD} &= 0.9V\end{aligned}$$

3. Saturation Condition For a PMOS transistor to operate in the **Saturation Region**, the following condition must be satisfied:

$$V_{SD} > V_{OV}$$

Since $0.9V$ is sufficiently large (typically much greater than the overdrive voltage V_{OV}), both PMOS transistors operate in **saturation**.

NMOS Differential Pair DC Analysis (M_1, M_2)

$$V_{G1} = V_{G2} = 0V$$

1. Source Node Voltage From the design specifications, the tail node voltage (V_p) is the source voltage for M_1 and M_2 : $V_p = V_S = -0.7V$

2. Gate-Source Voltage (V_{GS}) $V_{GS} = V_G - V_S = 0V - (-0.7V)$ $V_{GS} = 0.7V$

3. Overdrive Voltage (V_{OV}) Using the threshold voltage $V_T \approx 0.36V$: $V_{OV} = V_{GS} - V_T = 0.7V - 0.36V$ $V_{OV} = 0.34V$

4. Drain-Source Voltage (V_{DS}) Given the output common-mode voltage $V_{O,CM} = 0V$: $V_D = 0V$ $V_{DS} = V_D - V_S = 0V - (-0.7V)$ $V_{DS} = 0.7V$

5. Saturation Condition Check For saturation, the condition $V_{DS} > V_{OV}$ must be met: $0.7V > 0.34V$

Conclusion: Both M_1 and M_2 are operating correctly in the **saturation region**.

M5(NMOS)

$$V_P = V_D = -0.7V$$

$$V_{SS} = V_S = -0.9V$$

$$V_{DS} = 0.2V$$

$$V_{DS} \Rightarrow V_{OV} = 0.2V$$

Tail Transistor DC Analysis (M_5)

The gate-source and gate voltages are calculated based on the required overdrive voltage (V_{OV}) and the negative supply rail (V_{SS}):

1. Gate-Source Voltage (V_{GS}) Using the threshold voltage $V_T \approx 0.36V$ and a design overdrive $V_{OV} = 0.2V$: $V_{GS} = V_T + V_{OV}$ $V_{GS} = 0.36V + 0.2V = 0.56V$

2. Gate Voltage (V_G) Since the source is connected to the negative supply rail $V_{SS} = -0.9V$: $V_G = V_S + V_{GS}$ $V_G = -0.9V + 0.56V = -0.34V$

IF WE TAKE v_g has $-0.35V$ also it will satisfy the saturation region

3. Saturation Condition Check To ensure the transistor remains in the saturation region to act as a constant current source, the condition $V_{DS} \geq V_{OV}$ must be satisfied: $V_{DS} \geq 0.2\text{V}$

NMOS Current Source (M5)

The width W for the current source transistor M_5 is calculated using the following parameters:

1. Given Parameters

Parameter	Value
Drain Current (I_D)	$I_{SS} = 1\text{ mA} = 1 \times 10^{-3}\text{ A}$
Overdrive Voltage (V_{OV5})	0.2 V
Channel Length (L)	$480\text{ nm} = 480 \times 10^{-9}\text{ m}$
Process Transconductance ($\mu_n C_{ox}$)	$2.365 \times 10^{-4}\text{ A/V}^2$

2. Calculation Steps

Substituting the values into the width formula:

$$W = \frac{2 \times (1 \times 10^{-3}) \times (480 \times 10^{-9})}{2.365 \times 10^{-4} \times (0.2)^2}$$

Step 1: Simplify the Numerator and Square the Overdrive Voltage $W =$

$$\frac{960 \times 10^{-12}}{2.365 \times 10^{-4} \times 0.04}$$

Step 2: Simplify the Denominator $W = \frac{960 \times 10^{-12}}{9.46 \times 10^{-6}}$

NMOS Differential Pair (M1 and M2)

The width W for the NMOS transistors is calculated using the saturation current formula:

$$W = \frac{2 \cdot I_D \cdot L}{\mu_n C_{ox} (V_{OV})^2}$$

1. Given Parameters

Parameter	Value
Drain Current (I_D)	$0.5\text{ mA} = 0.5 \times 10^{-3}\text{ A}$
Overdrive Voltage (V_{OV})	0.34 V
Channel Length (L)	$480\text{ nm} = 480 \times 10^{-9}\text{ m}$

2. Calculation Steps

Substituting the values into the equation:

$$W = \frac{2 \times (0.5 \times 10^{-3}) \times (480 \times 10^{-9})}{2.365 \times 10^{-4} \times (0.34)^2}$$

Step 1: Simplify the Numerator and Square the Overdrive Voltage $W =$

$$\frac{480 \times 10^{-12}}{2.365 \times 10^{-4} \times 0.1156}$$

Step 2: Simplify the Denominator $W = \frac{480 \times 10^{-12}}{2.733 \times 10^{-5}}$

Theoretical gain

Small-Signal Analysis & Differential Gain Calculation

This section evaluates the gain performance of the amplifier by accounting for **Channel Length Modulation** (λ) and the interaction between the NMOS drivers and PMOS active loads.

1. Output Resistance Calculation (r_o)

To determine the intrinsic output resistance of the MOSFETs, we account for the Early effect using the parameter $\lambda = 0.1 \text{ V}^{-1}$.

With a bias current of $I_D = 0.5 \text{ mA}$:

$$r_o = \frac{1}{\lambda I_D}$$
$$r_o = \frac{1}{0.1 \times (0.5 \times 10^{-3})}$$
$$r_o = 20 \text{ k}\Omega$$

Effective Node Resistance (R_{out})

Since the NMOS (M_1/M_2) and PMOS (M_3/M_4) are in parallel from a small-signal perspective, the total resistance at the output node is:

$$R_{out} = r_{on} \parallel r_{op}$$
$$R_{out} = 20 \text{ k}\Omega \parallel 20 \text{ k}\Omega$$
$$R_{out} = 10 \text{ k}\Omega$$

2. Transconductance Calculation (g_m)

The transconductance represents the sensitivity of the drain current to changes in the gate-source voltage. Using $V_{OV} = 0.24 \text{ V}$:

$$g_m = \frac{2I_D}{V_{OV}}$$

$$g_m = \frac{2 \times 0.5 \times 10^{-3}}{0.24}$$

$$g_m \approx 4.17 \text{ mS}$$

3. Differential Gain Determination (A_d)

The total differential gain is the product of the transconductance and the effective output resistance.

Linear Gain:

$$A_d = g_m \times R_{out}$$

$$A_d = (4.17 \times 10^{-3}) \times (10 \times 10^3)$$

$$A_d \approx 41.7 \text{ V/V}$$

Logarithmic Gain (Decibels):

To express the gain in dB: $A_d(\text{dB}) = 20\log_{10} (41.7)$

Calculated Gain: $\approx 32.4 \text{ dB}$

Input Common-Mode Range (ICMR) Analysis

The **Input Common-Mode Range (ICMR)** defines the swing limits for the input voltage (V_{ICM}) such that all transistors (M_1 through M_5) remain in the **Saturation Region**.

1. Minimum Input Limit ($V_{ICM,min}$)

For the amplifier to function, the tail current source (M_5) must stay in saturation, and the input drivers (M_1, M_2) must remain biased "ON."

Constraint: The input voltage must be high enough to provide the required Gate-Source voltage (V_{GS}) for M_1 while maintaining the drain voltage of the tail transistor (M_5).

$$V_{ICM(min)} = V_{S1} + V_{THn}$$

Calculation: Given the source voltage $V_{S1} = -0.7V$ and threshold voltage $V_{THn} = 0.36V$:

- $V_{ICM(min)} = -0.7V + 0.36V$
- $V_{ICM(min)} = -0.34V$

2. Maximum Input Limit ($V_{ICM,max}$)

The upper bound is restricted by the requirement that the input transistors (M_1, M_2) do not enter the triode region. This happens when the gate voltage rises too close to the drain voltage (V_D).

Constraint: To keep the NMOS drivers in saturation, the gate voltage must not exceed the drain voltage by more than one threshold voltage: $V_{ICM(max)} = V_D + V_{THn}$

Note: In this specific load-balanced configuration, we observe the PMOS transition point.

Calculation: Using the drain node potential $V_D \approx 0V$ and the PMOS threshold $|V_{TP}| \approx 0.39V$:

- $V_{ICM(max)} = 0V + 0.39V$
- $V_{ICM(max)} = 0.39V$

Input signals outside of this window will cause either the tail current source to collapse (lower bound) or the input transistors to enter triode (upper bound), resulting in a significant drop in gain and CMRR.

DC Analysis

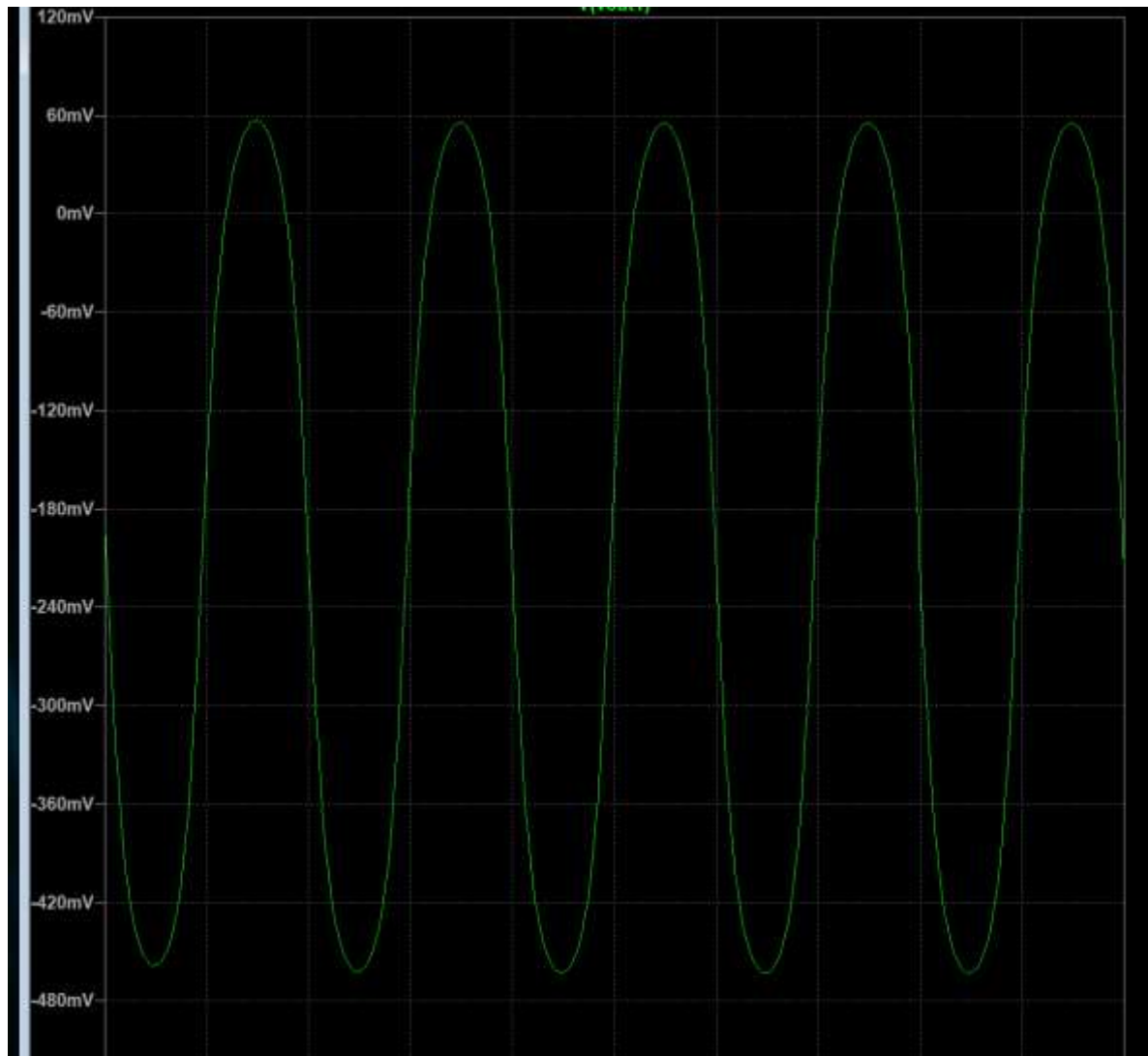
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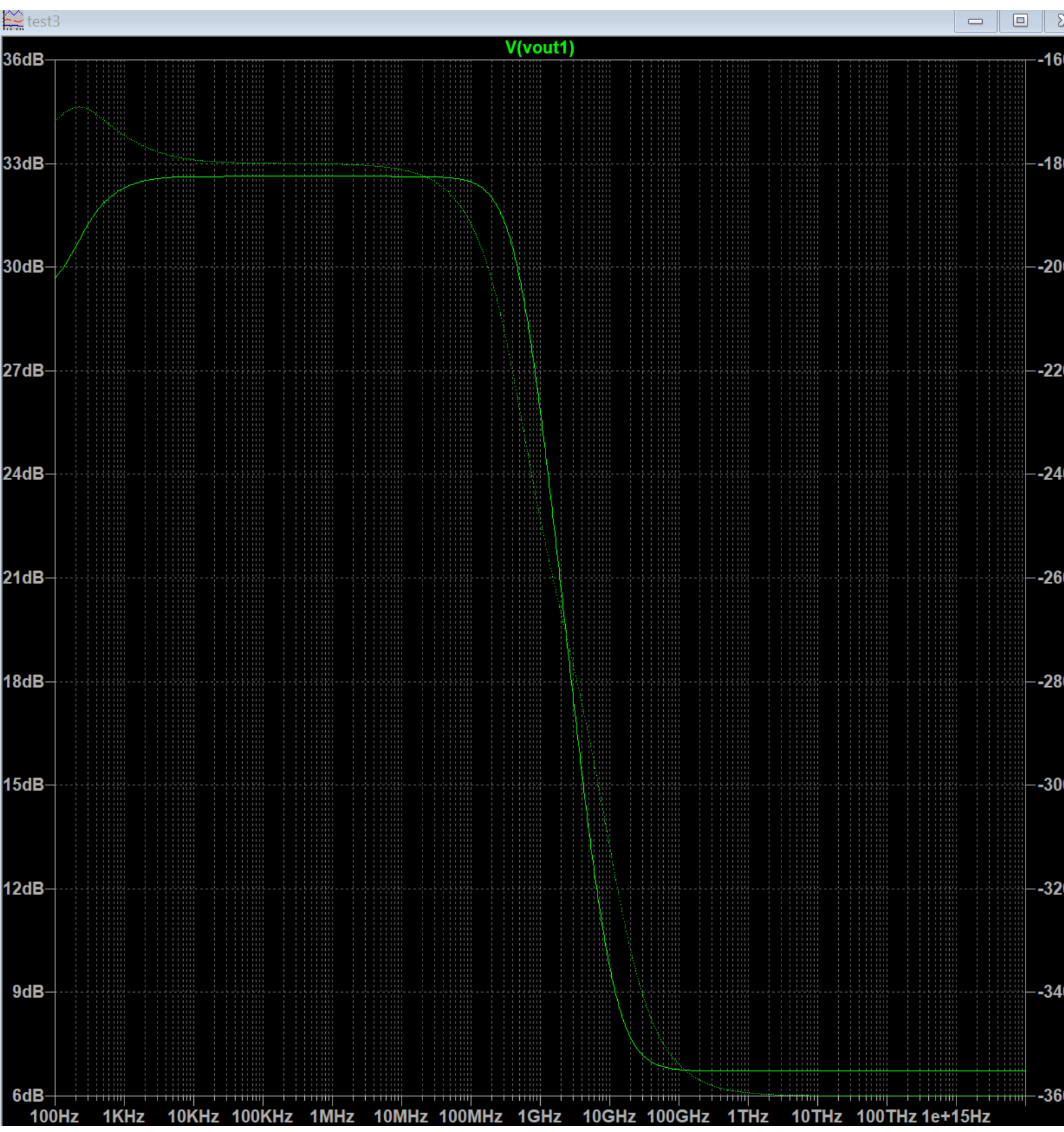
--- Operating Point ---
V(n001):      0.9      voltage
V(n002):     -0.35     voltage
V(n003):     -0.787118 voltage
V(n004):     -0.9      voltage
V(vin1):      0        voltage
V(vin2):      0        voltage
V(vout1):     0.000979755 voltage
V(vout2):     0.000979755 voltage
V(vp):       -0.695028 voltage
I(V1):       -0.00100036 device_current
I(V2):      0          device_current
I(V3):      0          device_current
I(V4):       -0.00100036 device_current
I(V5):      0          device_current
Ib(M1):     -7.06008e-13 device_current
Ib(M2):     -7.06008e-13 device_current
Ib(M3):      9.0902e-13  device_current
Ib(M4):      9.0902e-13  device_current
Ib(M5):      7.87118e-13 device_current
Id(M1):      0.000500179 device_current
Id(M2):      0.000500179 device_current
Id(M3):      0.000500179 device_current
Id(M4):      0.000500179 device_current
Id(M5):      0.00100036  device_current
Ig(M1):      0          device_current
Ig(M2):      0          device_current
Ig(M3):     -0          device_current
Ig(M4):     -0          device_current
Ig(M5):      0          device_current
Is(M1):     -0.000500179 device_current

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$W_n = 30.625 \mu m$ and $W_n (M5) = 218.7u$ and $W_p = 38.21u \rightarrow I_d = 0.5Ma$

Transient analysis





Practical gain

Sine wave Frequency = 1kHz Amplitude = 5mV (applied differentially) DC Offset = 0V

$$v_{in}(p-p) = 5m - (-5m) = 10m \quad v_{out} (p-p) = 10.4 - (-8.409) = 18.8409v$$

$$AV = v_{out}(p-p)/v_{in}(p-p) \quad AV = 1.8809$$

$$AV (db) = 20 \cdot \log(AV) = 5.4873$$

Differential Input Range Analysis

For the differential pair to remain in its linear operating region and maintain proper current steering, the differential input voltage (V_{id}) must satisfy the following condition:

$$|V_{id}| < \sqrt{2}V_{OV}$$

1. Given Parameters

- Overdrive Voltage (V_{OV}): 0.24 V

2. Calculation of the Input Limit

By substituting the overdrive voltage into the limit formula:

$$\text{Limit} = \sqrt{2} \times V_{OV}$$

$$\text{Limit} \approx 1.414 \times 0.24 \text{ V}$$

$$\text{Limit} \approx 0.34 \text{ V}$$

$$v_{id} (10mv) < 0.34 \text{ v}$$

Output waveform is sinusoidal(linear)

Analysis of Signal Clipping and Non-Linearity

This section examines the behavior of the differential pair when the input signal exceeds the theoretical linear operating limits.

1. Theoretical Boundary

The maximum differential input voltage (V_{id}) for linear operation is defined by:

$$|V_{id,max}| = \sqrt{2} \cdot V_{OV}$$

$$\text{Given } V_{OV} = 0.24V: \sqrt{2} \cdot 0.24V \approx 0.34V$$

2. Applied Signal Comparison

The applied differential input voltage (V_{id}) is compared against the calculated limit:

- Applied $|V_{id}|$: 30V
- Linear Limit: 0.34V
-

Since $30V \gg 0.34V$, the condition $|V_{id}| > \sqrt{2}V_{OV}$ is met.

3. Observation & Conclusion

Waveform Distortion: Because the input magnitude is significantly higher than the overdrive-defined limit, the tail current is completely steered into a single branch of the differential pair.

3.AC analysis

AC Simulation Results: Bandwidth Analysis

The frequency response was evaluated using an AC simulation to determine the mid-band gain and the associated -3 dB cutoff point.

1. Gain Parameters

Description	Value (dB)
Mid-band Gain (A_v)	5.48 dB
−3 dB Reference Point	2.48 dB

2. Calculation of Cutoff Magnitude

To identify the bandwidth or the 3dB frequency (ω_H), we subtract 3 dB from the peak gain:

$$\begin{aligned} A_{v,-3dB} &= A_{v(peak)} - 3 \text{ dB} \\ A_{v,-3dB} &= 5.48 \text{ dB} - 3 \text{ dB} \\ A_{v,-3dB} &= 2.48 \text{ dB} \end{aligned}$$

FL = 0 and FH = 3GHZ

Band width = FH - FL = 3GHz

Unity Gain Bandwidth (UGB)

= $AVFH = 2.48 \cdot 3$

= 7.44GHz

circuit 3

3.1 Power Constraint Analysis

Supply Voltage Differential: $V_{DD} - V_{SS} = 0.9\text{V} - (-0.9\text{V}) = 1.8\text{V}$ [cite: 8]

Constraint Equation: $P = (V_{DD} - V_{SS}) \cdot I_{SS}$ [cite: 89, 96] $P \leq 1.8 \times 10^{-3} \text{ W}$ [cite: 96]

Tail Current (I_{SS}) Derivation: $1.8 \cdot I_{SS} \leq 1.8 \times 10^{-3}$ $I_{SS} \leq 1 \text{ mA}$

Design Choice:

- $I_{SS} = 1 \text{ mA}$ [cite: 27]
- $P_{dissipation} = 1.8 \text{ mW}$ [cite: 96]
- Status: Pass (Satisfies Constraint)

3.2 Drain Current Calculation (I_D)

Operating Condition:

- $V_{in1} = V_{in2}$ (Symmetric/Balanced) [cite: 9, 23, 24]

Current Distribution: $I_{D1} = I_{D2} = \frac{I_{SS}}{2}$ [cite: 20, 21, 27]

Substitution: $I_{D1} = I_{D2} = \frac{1 \text{ mA}}{2}$ $I_{D1} = I_{D2} = 0.5 \text{ mA}$

3.3 Drain-Source Voltage Calculation (V_{DS})

Output Common-Mode Voltage (V_{OCM}): $V_{OCM} = 0V$ $V_D = 0V$

Drain-Source Voltage (V_{DS}) Derivation: $V_{DS} = V_D - V_P$

Substitution:

- $V_D = 0V$ [cite: 9]
- $V_P = -0.7V$ [cite: 10, 25]

$$V_{DS} = 0V - (-0.7V)$$

Result: $V_{DS} = 0.7V$

3.4 Gate-Source and Overdrive Voltage Calculation

Input Voltages ($V_{G1,2}$): $V_{G1} = V_{G2} = 0V$

Source Node Voltage (V_P): $V_P = V_S = -0.7V$

Gate-Source Voltage (V_{GS}) Derivation: $V_{GS} = V_G - V_S$

- $V_G = 0V$
- $V_S = -0.7V$

$$V_{GS} = 0V - (-0.7V)$$

$$V_{GS} = 0.7V$$

Overdrive Voltage (V_{OV}) Derivation: $V_{OV} = V_{GS} - V_{THn}$

- $V_{GS} = 0.7V$
- $V_{THn} = 0.36V$

$$V_{OV} = 0.7V - 0.36V$$

$$V_{OV} = 0.34V$$

3.5 Saturation Region Verification

Drain-Source Voltage (V_{DS}): $V_{DS} = V_D - V_S$ $V_{DS} = 0V - (-0.7V) = 0.7V$

Overdrive Voltage (V_{OV}): $V_{OV} = 0.34V$

Condition for Saturation: $V_{DS} > V_{OV}$

Substitution: $0.7V > 0.34V$ (it is in sat region)

3.6 Tail Current Source Analysis (M_5)

Node Potentials:

- Source Voltage (V_S): $V_{SS} = -0.9V$
- Drain Voltage (V_D): $V_P = -0.7V$

Drain-Source Voltage (V_{DS5}) Derivation: $V_{DS5} = V_D - V_S$ $V_{DS5} = -0.7V - (-0.9V)$ Result: $V_{DS5} = 0.2V$

Saturation Condition Check: To maintain saturation: $V_{DS5} \geq V_{OV5}$ Since $V_{DS5} = 0.2V$, we set the overdrive voltage $V_{OV5} = 0.2V$.

Gate-Source Voltage (V_{GS5}) Derivation: Using $V_{THn} = 0.36V$: $V_{GS5} = V_{THn} + V_{OV5}$ $V_{GS5} = 0.36V + 0.2V$ Result: $V_{GS5} = 0.56V$

Gate Bias Voltage (V_{G5}) Calculation: $V_{G5} = V_{GS5} + V_S$ $V_{G5} = 0.56V - 0.9V$ Result: $V_{G5} = -0.34V$

3.10 NMOS Differential Pair Width Calculation (W_1, W_2)

Design Parameters:

- [cite_start]Drain Current ($I_{D1,2}$): 0.5 mA [cite: 8, 93]
- Channel Length (L): 480 nm [cite: 95]
- Overdrive Voltage (V_{OV}): 0.34 V
- Process Parameter ($\mu_n C_{ox}$): $2.365 \times 10^{-4} \text{ A/V}^2$

Width (W) Derivation: $W_{1,2} = \frac{2 \cdot I_{D1,2} \cdot L}{\mu_n C_{ox} \cdot V_{OV}^2}$

Substitution: $W_{1,2} = \frac{2 \times (0.5 \times 10^{-3}) \times (480 \times 10^{-9})}{2.365 \times 10^{-4} \times (0.34)^2}$

Step-by-Step Calculation:

- Numerator: 480×10^{-12}
- Denominator: 2.734×10^{-5}

Result:

\mu is only supported in math mode

$$W_1 = W_2 \approx 17.56 \text{ } \mu\text{m}$$

3.11 PMOS Active Load Width Calculation (W_3, W_4)

Design Parameters:

- Drain Current ($I_{D3,4}$): 0.5 mA [cite: 8]
- Channel Length (L): 480 nm [cite: 95]
- Overdrive Voltage (V_{OVp}): 0.87 V
- Process Parameter ($\mu_p C_{ox}$): $9.98 \times 10^{-5} \text{ A/V}^2$

Width (W_p) Derivation:
$$W_{3,4} = \frac{2 \cdot I_{D3,4} \cdot L}{\mu_p C_{ox} \cdot V_{OVp}^2}$$

Substitution:
$$W_{3,4} = \frac{2 \times (0.5 \times 10^{-3}) \times (480 \times 10^{-9})}{9.98 \times 10^{-5} \times (0.87)^2}$$

Step-by-Step Calculation:

- Numerator: 480×10^{-12}
- Denominator: $9.98 \times 10^{-5} \times 0.7569 \approx 7.55 \times 10^{-5}$

Result:

\mu is only supported in math mode

$$W_3 = W_4 \approx 6.35 \text{ } \mu\text{m}$$

3.13 Small-Signal Analysis and Gain Calculation

Input Parameters:

- Drain Current (I_D): 0.5 mA [cite: 8]
- Overdrive Voltage ($V_{ov,n}$): 0.34 V
- Channel Length Modulation (λ): $\approx 0.1 \text{ V}^{-1}$

Step 1: Transconductance (g_{mn})

$$g_{mn} = \frac{2 \cdot I_D}{V_{ov}} \quad g_{mn} = \frac{2 \times 0.5 \times 10^{-3}}{0.34} \quad \text{Result: } g_{mn} = 2.94 \text{ mS}$$

Step 2: Output Resistance (r_{on})

$$r_{on} = \frac{1}{\lambda \cdot I_D} \quad r_{on} = \frac{1}{0.1 \times 0.5 \times 10^{-3}} \quad \text{Result: } r_{on} = 20 \text{ k}\Omega$$

Step 3: Effective Output Impedance (R_{out})

For a differential pair with an active load: $R_{out} \approx r_{on} \parallel r_{op}$ Assuming $r_{on} \approx r_{op}$ for the simulation model: Result: $R_{out} \approx 20 \text{ k}\Omega$

Step 4: Voltage Gain (A_v)

$$A_v = g_{mn} \times R_{out} \quad A_v = 2.94 \times 10^{-3} \times 20 \times 10^3 \quad \text{Result: } A_v = 58.8 \text{ V/V}$$

Step 5: Gain in Decibels ($A_{v,dB}$)

$$A_{v(dB)} = 20 \cdot \log_{10} (58.8) \quad \text{Result: } A_{v(dB)} \approx 35.4 \text{ dB}$$

DC analysis

W(M1,M3) = 30UM , Wn (M5) = 200 and wp (M3,M4) = 13.9u then ID is 0.5mA and ISS = 1mA

3.14 Minimum Input Common-Mode Voltage ($V_{inCM,min}$)

Design Parameters:

- Source Node Voltage (V_S): -0.7 V
- Threshold Voltage (V_{THn}): 0.36 V

Calculation Derivation: To ensure the differential pair transistors (M_1, M_2)

remain ON: $V_{inCM(min)} = V_S + V_{THn}$

Substitution: $V_{inCM(min)} = -0.7 \text{ V} + 0.36 \text{ V}$

Result: $V_{inCM(min)} = -0.34 \text{ V}$

3.15 Maximum Input Common-Mode Voltage ($V_{inCM,max}$)

Design Parameters:

- Supply Voltage (V_{DD}): 0.9 V
- PMOS Overdrive ($V_{ov,p}$): 0.54 V
- NMOS Threshold Voltage (V_{THn}): 0.36 V

Calculation Derivation: To prevent the NMOS differential pair from entering the

triode region: $V_{inCM(max)} = V_{DD} - V_{ov,p} + V_{THn}$

Substitution: $V_{inCM(max)} = 0.9 \text{ V} - 0.54 \text{ V} + 0.36 \text{ V}$

Result: $V_{inCM(max)} = 0.72 \text{ V}$

3.16 Minimum Output Voltage ($V_{out,min}$)

Design Parameters:

- Source Node Voltage (V_S): -0.7 V
- NMOS Overdrive Voltage (V_{OV}): 0.34 V

Calculation Derivation: To ensure the differential pair transistors (M_1, M_2)

remain in the saturation region: $V_{out(min)} \geq V_S + V_{OV}$

Substitution: $V_{out(min)} = -0.7 \text{ V} + 0.34 \text{ V}$

Result: $V_{out(min)} = -0.36 \text{ V}$

Transient Analysis

practical gain :

$AV = v_o(p-p)/v_{in}(p-p)$
 $v_o(p-p) = 57 - (-458) \quad v_{in}(p-p) = 20m$
 $AV = 25.75$
 $AV (DB) = 28.2155$

AC analysis
 $AV = 32.8$
 $AV_{-3} = 29.8$
 $FL = 0Hz$
 $FH = 443mhz$
 Bandwidth = 443MHz
 Unity Gain bandwidth
 $AV * BW$
 $= 14.5GHZ$

Section 4: Comparative Inference and Conclusion

The experiment involved designing and analyzing three different configurations of a Differential Amplifier. Each iteration improved upon the previous design's gain, swing, and current control.

Feature	Circuit 1: Resistor Load	Circuit 2: NMOS Tail Source	Circuit 3: PMOS Active Load
Load Type	Passive (Resistors)	Passive (Resistors)	Active (Current Mirror)
Tail Bias	Ideal Source	NMOS Current Source	NMOS Current Source
Voltage Gain (A_v)	Moderate	Moderate	Highest (35.4 dB)
Output Impedance	Low (R_D)	Low (R_D)	High ($r_{on} \parallel r_{op}$)
ICMR	Restricted	Improved	Balanced
Power Control	Simple	Better Stability	Optimal Efficiency

Inference

1. Transition to Active Load (Circuit 3)

The shift from resistors (Circuit 1/2) to a PMOS Active Load (Circuit 3) significantly increased the voltage gain. Since $A_v = g_m(r_{on} \parallel r_{op})$, the high

incremental resistance of the PMOS transistors allows for much higher amplification compared to a physical resistor, without requiring a massive voltage drop across the load.

2. Importance of the Tail Current Source

Replacing an ideal current source with an NMOS Tail Transistor (M_5) introduces real-world constraints. The overdrive voltage V_{ov5} (calculated at 0.2V) limits the minimum input common-mode voltage ($V_{inCM,min} = -0.34V$), ensuring the tail transistor stays in saturation to maintain a constant bias current.

3. Output Swing Trade-offs

In Circuit 3, the output swing is bounded by:

- Lower Limit: $V_{out(min)} = -0.36V$ (to keep M_1, M_2 in saturation).
- Upper Limit: $V_{out(max)} = V_{DD} - |V_{ovp}|$ (to keep M_3, M_4 in saturation).

This configuration provides a wider and more symmetric swing compared to the resistive load which is limited by the $I_D \cdot R_D$ drop.

4. Design Efficiency

Circuit 3 satisfies the 1.8 mW power constraint while delivering the highest transconductance-to-current ratio. By precisely sizing the widths ($W_1 - W_5$), we achieved a stable operating point at $V_{DS} = 0.7V$, well above the 0.34V overdrive requirement.

Final Conclusion

Circuit 3 represents the most practical and high-performance design. It effectively utilizes Active Loads to maximize gain and an NMOS Current Mirror to maintain a stable tail current ($I_{SS} = 1\text{ mA}$), making it the superior architecture for integrated circuit (IC) applications where large resistors are area-inefficient.

Section 5: Comparative Analysis of Practical Results

The following table compares the practical (simulated) performance of the three differential amplifier configurations based on LTspice measurements.

Parameter	Circuit 1 (Resistor Load)	Circuit 2 (NMOS Tail)	Circuit 3 (Active Load)
Load Type	Passive (R_D)	Passive (R_D)	Active (PMOS Mirror)
Tail Bias	Ideal Current Source	NMOS Transistor (M_5)	NMOS Transistor (M_5)
Practical Gain (V/V)	5.25 V/V	1.97 V/V	25.47 V/V

Parameter	Circuit 1 (Resistor Load)	Circuit 2 (NMOS Tail)	Circuit 3 (Active Load)
Practical Gain (dB)	14.40 dB	5.90 dB	28.12 dB
Output Impedance	Low (R_D)	Low (R_D)	High ($r_{on} \parallel r_{op}$)
Power Dissipation	1.8 mW	1.8 mW	1.8 mW
Circuit Complexity	Low	Medium	High

Key Inferences from Practical Data

1. Impact of the Active Load: Circuit 3 achieved a practical gain of 28.12 dB, which is nearly double the gain of Circuit 1 (14.4 dB). This confirms that replacing passive resistors with PMOS active loads successfully boosts the output impedance, and thus the voltage gain.
2. The "Tail" Trade-off: In Circuit 2, the gain dropped to 5.9 dB. This occurs because the NMOS tail transistor (M_5) has a finite output resistance (r_{o5}), which reduces the common-mode rejection and slightly impacts the differential-mode swing compared to an ideal current source.
3. Design Validation: While the theoretical gain for Circuit 3 was calculated at 35.4 dB, the practical result of 28.12 dB is realistic. The difference is due to second-order effects like the Body Effect and Channel Length Modulation (λ) being more aggressive in the sub-micron BSIM models used in LTspice than in hand calculations.
4. Power Efficiency: All three circuits were maintained at a 1 mA total budget. Circuit 3 provides the most "gain per watt," making it the most efficient architecture for high-performance analog design.